



Documentation Quality Compliance Checker

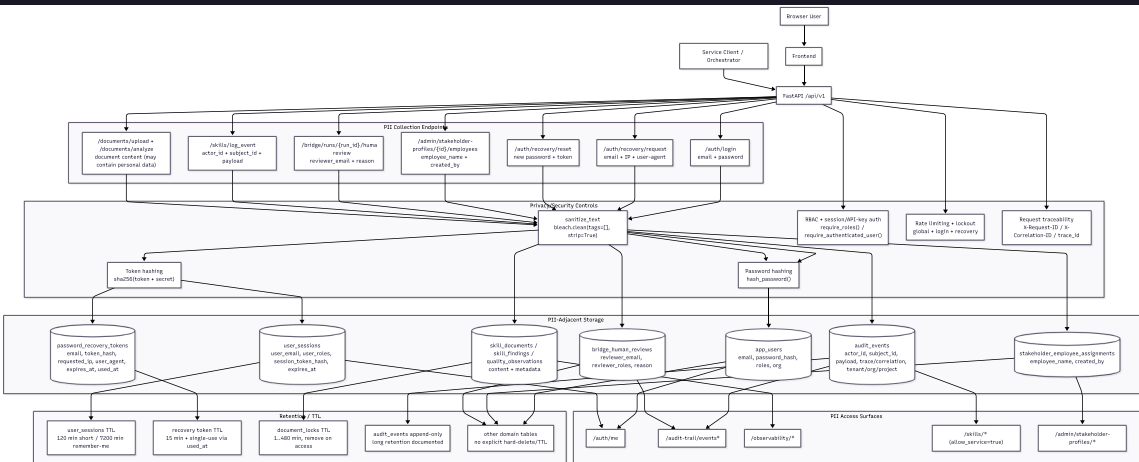
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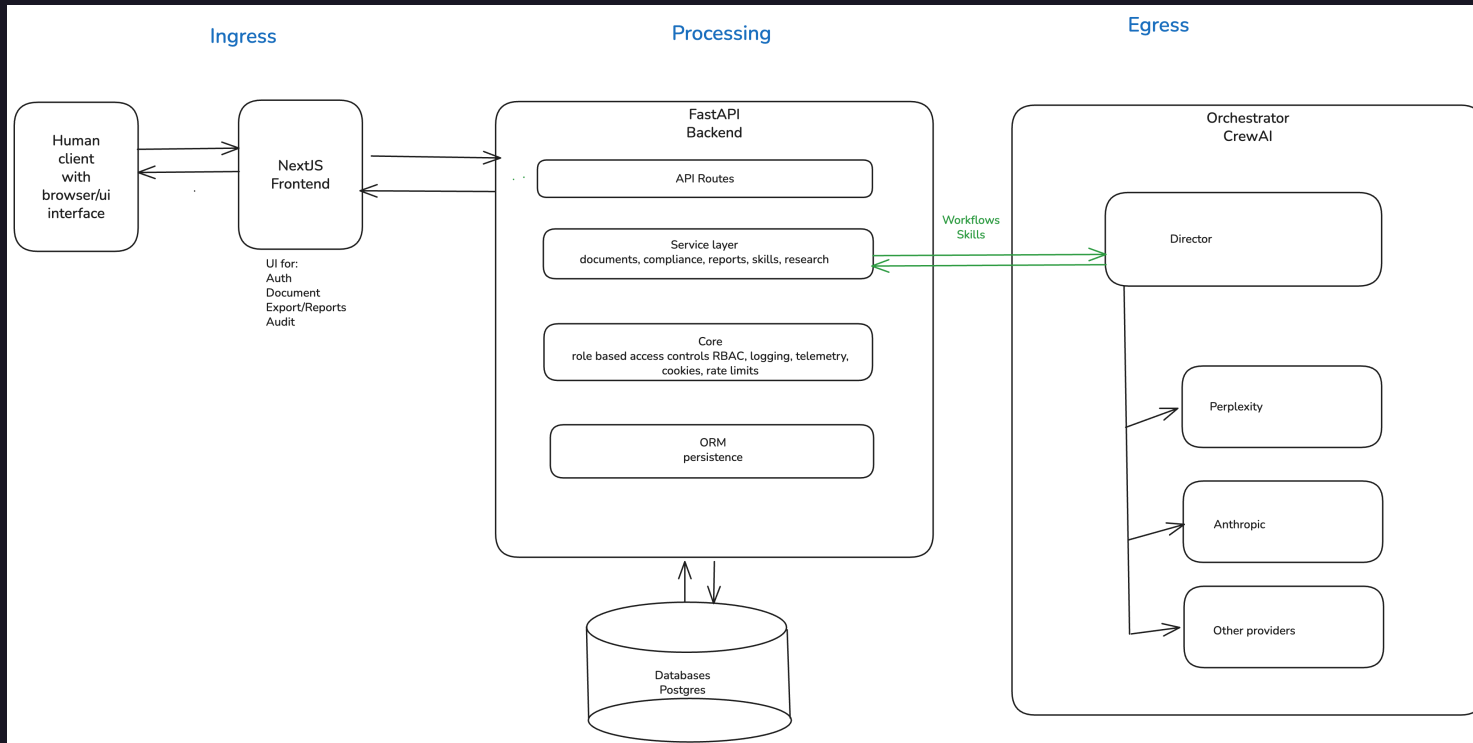
Project & use case

- What we audited
- Problem solved
- Architecture at a glance
- Why privacy matters

What we audited

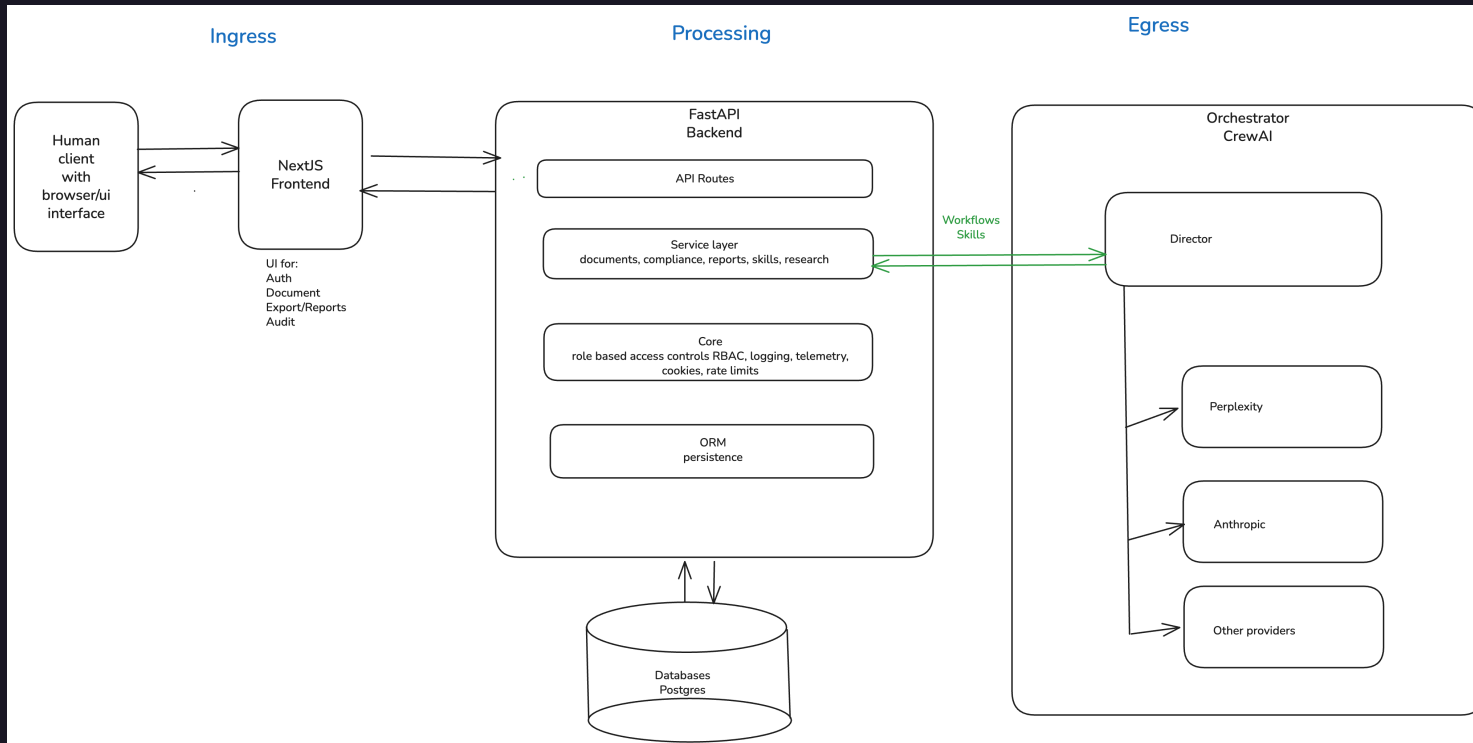
- **Scope:** Documentation quality compliance checker performs AI-assisted QM audits of technical docs
- **Repo:** https://github.com/lloBe/doc_quality_compliance_checker
- **Audit:** <https://willingc.github.io/compliance-checker/>





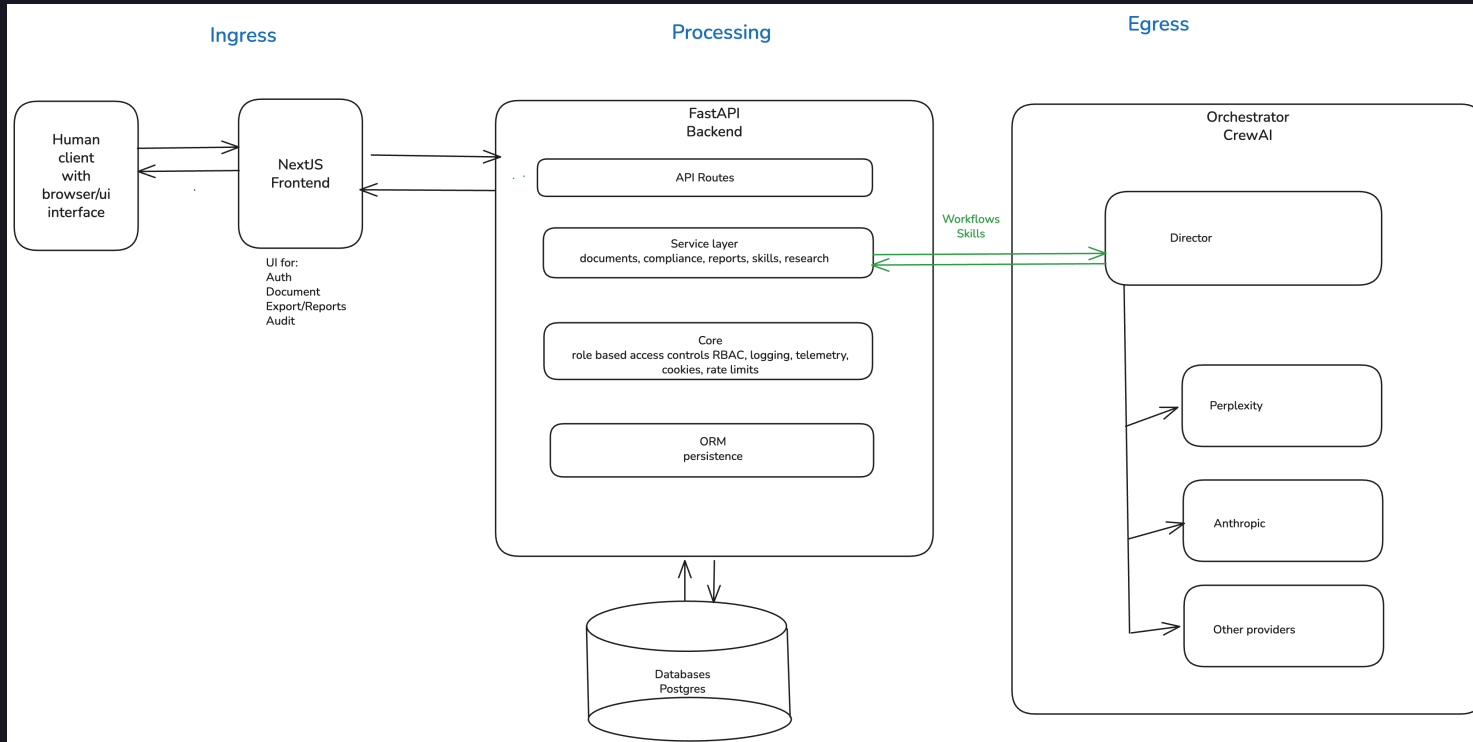
Problem solved

- Teams must prove documentation meets quality and regulatory expectations
- Manual QM review is slow, inconsistent, and hard to scale
- AI agents can help — but introduce new privacy and data-flow risks



High-level architecture

- Human operator
- Next.js frontend
- FastAPI backend
- PostgreSQL database
- Orchestrator and agents



Privacy matters

- User accounts, roles, and session data (GDPR-relevant)
- Document content and audit trails may contain sensitive business data
- Egress to LLM providers and telemetry expand the trust boundary

Privacy risks

- Top privacy risks
- Risk areas in depth
- Prioritization and trade-offs

2.0

Top privacy risks

1. Egress traffic flow data to orchestrator and models
2. Application telemetry and workflow tracing
3. Role-based access controls and GDPR compliance

2.1 Risk Area 1

Egress traffic flow data to orchestrator and models

Sensitive data:

- External model: transfer of potentially personal prompt/output data
 - **Data:**
 - Names, emails, stakeholder assignments, reviewer identifiers, document passages copied into prompts, generated summaries that may restate personal data
 - **Why Sensitive:**
 - Leaves the primary application context during model inference (current external-provider path)

- Over-collection and -retention in model observability traces
 - **Data:**
 - provider, model_used, trace_id, correlation_id, latency/tokens, rich payload entries, prompt/output snapshots
 - **Why Sensitive:**
 - Enables reconstruction of *who* triggered *what* model action and *when*
 - can become re-identifiable when combined with audit tables

- Model credentials and routing configuration
 - **Data:**
 - API keys, adapter routing flags, provider selection settings
 - **Why Sensitive:**
 - Compromise enables data exfiltration or unauthorized model usage

2.2 Risk Area 2

Application telemetry and workflow tracing

Sensitive data:

- Raw LLM prompt and output in *quality_observations.payload*
 - **Data:**
 - *payload.provider, payload.model_used, payload.llm_temperature*
 - *payload.llm_prompt* (document text submitted by user, stakeholder names, reviewer assignments)
 - *payload.llm_output* (generated compliance summaries restating personal context)

- ○ **Why Sensitive:**
 - Prompt context is assembled from user-submitted documents which routinely contain names, emails, project identifiers, and business-sensitive content
 - The output may restate or summarise that content
 - Both are persisted to a queryable table

- Audit event payload in `audits_events.payload`
 - **Data:**
 - `payload.roles` (user role at login), `payload.remember_me` flag, event-specific context fields, any free-form data inserted by orchestrator or Skills API callers
 - **Why Sensitive:**
 - Append-only and intended for long retention
 - No technical control prevents a caller from writing personal data into the payload column
 - Combined with `actor_id` (user email) it forms a rich personal profile

Risk Area 2

- OTEL (open telemetry) span attributes (exporter config)
 - **Data:**
 - HTTP path attribute (e.g., `/api/v1/documents/doc-abc123` — document ID in path), `user_agent`, `http.method`, `http.status_code`, `trace_id` / `span_id` (linkable back to session)
 - **Why Sensitive:**
 - When `TRACING_EXPORTER=otlp` span data is sent to an external collector,
 - path values can embed document or session identifiers
 - user-agent can contribute to fingerprinting

Risk Area 2

- Frontend CSV export of prompt/output pairs
 - **Data:**
 - Exported observability *prompt_output_pairs.csv* file containing prompt, output, source_component, trace_id, timestamps
 - **Why Sensitive:**
 - Created on the user's local filesystem;
 - outside system access controls, retention policy, and audit log;
 - may contain personal data from documents processed during the export window

2.3 Risk Area 3

Role-based access controls and GDPR compliance

Sensitive data:

- User identity in session store
 - **Data:**
 - user_email (primary identifier), user_org, session session_id, expires_at, last_seen_at
 - **Why Sensitive:**
 - Directly identifies users and links their organisational role to access patterns and audit trails
 - Retained in PostgreSQL with no visible TTL-based purge policy

- Role assignments and permission scope
 - **Data:**
 - user_roles array per session row (e.g., ["qm_lead"], ["auditor"]), org isolation field user_org
 - **Why Sensitive:**
 - Reveals organisational responsibilities and access privileges
 - Can be used for social engineering or targeted attacks
 - GDPR data minimisation applies

- Bootstrap/MVP credentials in environment configuration
 - **Data:**
 - AUTH_MVP_EMAIL, AUTH_MVP_PASSWORD, AUTH_MVP_ROLES, AUTH_MVP_ORG (env vars)
 - SECRET_KEY (API key secret)
 - **Why sensitive:**
 - Compromise of bootstrap credentials gives attacker a fully provisioned account with configurable roles
 - SECRET_KEY grants service-client access to skills and observability

- Access decision and audit context not separately persisted
 - **Data:**
 - Which role accessed which route at what time
 - 403 access denials
 - service-client route usage with payload summary
 - **Why sensitive:**
 - Absence of access-decision audit log prevents retrospective investigation of data-access incidents
 - Required for GDPR Art. 30 record of processing activities and breach response

Mitigations & controls

- Mitigation per top risk
- Technical approach and trade-offs
- Outcomes and learnings

3.0

Mitigations and controls

- Mitigation implemented
- Technical details
- Selected approach vs. alternatives
- Outcome and learnings

3.1 Mitigating Risk 1

Egress traffic flow data to orchestrator and models

3.2 Mitigating Risk 2

Application telemetry and workflow tracing

3.3 Mitigating Risk 3

Role-based access controls and GDPR compliance

Recommendations

- Top priority going forward
 - Why it matters
- Fit with company and risk strategy
 - Value created

Top priority going forward

- Insert what the top priority is

Why it matters

- Insert why it is important (ROI, risk reduction, user impact, compliance, security)

Fit with company and risk strategy

- Insert how it fits the broader company and risk strategy

Value created

- Insert what value it creates for the organization

Questions & answers

- Audit scope and findings
- Risk prioritization and trade-offs
- Mitigations and open items

Thank you

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Privacy audit

<https://willingc.github.io/compliance-checker/>

